

MONTHLY DEPARTMENTAL PROGRESS REPORT

Department of Computer Science & Engineering

Reporting Period: April 2026 • Date of Submission: 25/04/2026

REPORTING OVERVIEW & DEPARTMENT PROFILE

Department Name	Department of Computer Science & Engineering
Head of Department (HOD)	Dr. Kethineni Vinod Kumar
Reporting Cycle Period	April 2026
Date of Submission	25/04/2026
Activities Summary	35 recorded activities • 4 Publications • 4 Syllabus Course(s)

1A. JOURNAL PUBLICATIONS (2)

- 01 Deep Transfer Learning Architecture for Automated Early Detection of Retinal Pathologies in Fundus Imagery**
Author(s): Dr. Kethineni Vinod Kumar, Dr. P. Ramanathan • **Journal:** IEEE Transactions on Artificial Intelligence in Medicine
ISSN/ISBN: 2691-4581 **Issue:** 15/3 **Pages:** 112-126 **Indexed in:** IEEE / SCI / Scopus
Paper Link / DOI: <https://doi.org/10.1109/TAIM.2026.3129482>
- 02 Federated Edge Learning with Differential Privacy Guarantees for Autonomous Vehicular Ad-Hoc Networks**
Author(s): Er. M. Sateesh, Dr. Kethineni Vinod Kumar • **Journal:** ACM Transactions on Cyber-Physical Systems
ISSN/ISBN: 2378-962X **Issue:** 10/2 **Pages:** 45-61 **Indexed in:** ACM / Scopus
Paper Link / DOI: <https://doi.org/10.1145/3641209>

1B. CONFERENCE PRESENTATIONS (2)

- 01 Zero-Knowledge Proof Based Authentication Protocol for Decentralized Multi-Cloud Identity Management**
Author(s): Dr. Kethineni Vinod Kumar, K. Likhitha • **Conference:** IEEE International Conference on Cloud Computing (IEEE CLOUD 2026)
Date: 28th April 2026 **Location/Mode:** San Diego, USA (Hybrid) **Indexing:** IEEE Xplore
Proceedings Link: <https://ieee-cloud2026.org/proceedings/cse-019>
- 02 Benchmarking LLM Hallucination Mitigation Techniques in Biomedical Clinical Summarization**
Author(s): Er. C. Harika, Dr. Kethineni Vinod Kumar • **Conference:** International Conference on Machine Learning & Natural Language (ICMLNL 2026)
Date: 12th April 2026 **Location/Mode:** Hyderabad, India (In-Person) **Indexing:** Springer LNCS
Proceedings Link: <https://springer.com/conf/icmlnl2026/paper77>

1C. PATENTS (2)

- 01 Decentralized Blockchain-Enabled Dynamic Consensus Architecture for Secure Electronic Health Records**
Inventors: Dr. Kethineni Vinod Kumar, Er. M. Sateesh • **Applicants:** Sanskriti School of Engineering
Patent No: 202641029184 A | **Status:** Published | **Date:** 08/04/2026
- 02 Autonomous Low-Latency Drone Navigation System Using Edge Spiking Neural Networks**
Inventors: Dr. Kethineni Vinod Kumar, Dr. V. Annapurna • **Applicants:** Sanskriti School of Engineering
Patent No: 202541088219 B | **Status:** Awarded / Granted | **Date:** 02/04/2026

1D. ENTREPRENEURSHIP & START-UP INITIATIVES (2)

01 Smart Campus AI & GenAI Hackathon: Building Generative Assistance for Students

Date: 06/04/2026 | Participants: 120

Key Outcomes: 4 MVP prototypes developed; 2 teams selected for NASSCOM incubator seed funding

02 Web3 & Decentralized App Incubation Pitch for FinTech Solutions

Date: 18/04/2026 | Participants: 45

Key Outcomes: 1 student start-up (ChainGuard AI) secured angel funding commitment of Rs. 2 Lakhs

2. NSS & OTHER EXTENSION ACTIVITIES (2)

01 Rural Digital Literacy & Cyber Safety Awareness Campaign in Government Schools

Date: 11/04/2026 | Venue: Zilla Parishad High School, Beedupalli | Participants: 65 Student Volunteers

Outcomes: Trained 180 rural high school students on cyber scam awareness, UPI security, and basic coding

02 E-Waste Segregation and Responsible Recycling Drive

Date: 22/04/2026 | Venue: SSE Knowledge Park Campus | Participants: 50 Volunteers

Outcomes: Collected 210 kg of obsolete computer peripherals and handed over to certified green recycler

3. FACULTY DEVELOPMENT PROGRAMS (FDPS) (2)

01 One-Week National Level FDP on Generative AI, LLMOps and Foundation Models in Production

Organizing Body: IIT Hyderabad & SSE Department of CSE • **Dates:** 13/04/2026 to 18/04/2026

Role: Organizing Chair & Participant (Hybrid Mode) | **Key Outcomes:** Faculty mastered fine-tuning LLaMA-3 models, LangChain agent workflows, and vector databases

02 AICTE ATAL FDP on Cloud-Native Microservices Architecture with Kubernetes & Docker

Organizing Body: NIT Warangal • **Dates:** 07/04/2026 to 11/04/2026

Role: Participant (Online Mode) | **Key Outcomes:** Trained on production container orchestration, CI/CD pipeline automation, and service mesh

4. STUDENT DEVELOPMENT PROGRAMS (SDPS) (2)

01 Full-Stack Web Development Mastery: MERN Stack & Next.js 14 Hands-on Bootcamp

Date: 15/04/2026 to 17/04/2026 • **Resource Person:** Mr. K. Anirudh (Senior Software Architect, Atlassian) • **Participants:** 85

Outcomes: 85 students built and deployed live e-commerce portals on Vercel with MongoDB Atlas

02 Hands-on Cyber Security & Ethical Hacking Crash Course on Penetration Testing

Date: 24/04/2026 • **Resource Person:** Mr. V. Sandeep (Certified Ethical Hacker & Security Analyst, CyberDef) • **Participants:** 65

Outcomes: 65 students practiced vulnerability assessment using Kali Linux, Wireshark, and Burp Suite

5. HONORS, AWARDS & CERTIFICATIONS

Faculty Honor: Dr. Kethineni Vinod Kumar — Distinguished AI Researcher & Educator of the Year Award (Computer Society of India (CSI) - AP State Chapter, 19/04/2026)

Faculty Honor: Er. C. Harika — Best Technical Paper Award in Natural Language Processing (International Conference on Smart Computing (ICSC 2026), 08/04/2026)

Student Honor: Y. Sai Charan (22SSE1A0582) & Team Alpha — 1st Prize & Rs. 1,00,000 Cash Award in Smart India Hackathon (SIH 2026) at Smart India Hackathon 2026 Grand Finale (Ministry of Education (MoE) & AICTE, 15/04/2026 to 16/04/2026)

Student Honor: K. Divya (23SSE1A0539) — Gold Medal in State-Level Coding Olympiad & AlgoSprint at CodeKshatra 2026 (JNTU Anantapur, 21/04/2026)

Certification: AWS Certified Solutions Architect - Associate (Global Cloud Certification) on Amazon Web Services (AWS) Academy — Certified: 48 Certified

Certification: Google Cloud Professional Data Engineer Certification (Industry Certification) on Google Cloud Skills Boost — Certified: 29 Certified

Certification: Deep Learning Specialization with PyTorch & TensorFlow (DeepLearning.AI Professional Certificate) on Coursera / DeepLearning.AI — Certified: 54 Certified

6A. DEPARTMENT MEETINGS & ACADEMIC GOVERNANCE (2)

01 Department Meeting #1

Date: 04.04.2026

Decisions & Key Discussions: Review of Capstone Project external viva schedule, allocation of faculty mentors for Smart India Hackathon, and academic audit of lab records.

Policy Updates: Mandatory Git repository submission and automated code quality linting for all major projects.

Minutes Link: <https://sseptp.org/iqac/cse/minutes-04-04-2026>

02 Department Meeting #2

Date: 20.04.2026

Decisions & Key Discussions: Analysis of Mid-term examinations, syllabus coverage review across all 4 years, and organization of CYBERKSHETRA 2026 annual symposium.

Policy Updates: Minimum 90% attendance mandatory for appearing in pre-final lab practical exams.

Minutes Link: <https://sseptp.org/iqac/cse/minutes-20-04-2026>

6B. COLLABORATIONS & MOUS (2)

1. Infosys Springboard & SSE Department of Computer Science & Engineering • Period: Valid 2024 to 2027 (3 Years) • Scope: Digital Learning Curriculum, Industry Live Projects, Faculty Enablement, and Campus Placement Support

2. Red Hat Academy Center of Excellence • Period: Valid 2025 to 2028 (3 Years) • Scope: Red Hat Enterprise Linux (RHEL) Certification, OpenShift Cloud Native Lab, and Student Apprenticeship

7-9. TECHNICAL ASSOCIATIONS & INNOVATION INITIATIVES

Additional Initiative: 24/7 Competitive Coding Club & LeetCode Student Leaderboard (02/04/2026) — Department launched daily automated coding challenges with weekly faculty reviews on data structures & algorithms.

Additional Initiative: Open Source Software Contribution Sprint for Linux Kernel & Apache Projects (16/04/2026) — Weekend sprint guiding students through Git pull requests, open-source documentation, and code reviews.

Technical Association: CYBERKSHETRA 2026 - Annual National Technical Symposium & Hack-Fest (23/04/2026) — Conducted live bug-hunting tournament, paper presentations, and algorithmic coding championship.

Technical Association: Masterclass on Cloud Microservices & Scalable System Design (10/04/2026) — In-depth architectural analysis of distributed caching, Kafka event streaming, and sharding.

IIC Cell: GenAI & Cloud Innovation Pitch Sprint for Smart Governance Applications (14/04/2026) — Under IIC 6.0, students pitched AI solutions for public grievances and crop disease surveillance.

IIC Cell: Workshop on Patent Search & Software Copyright Drafting for Computer Innovations (25/04/2026) — Practical training on novelty claims for algorithmic inventions and Indian Patent Office software guidelines.

10. SYLLABUS COVERAGE & CURRICULUM PROGRESS (4)

S.No	Subject / Course	Year/Sem	Faculty	% Done	% Pend	Remarks
1	Operating Systems & System Programming	II B.Tech II Sem	Dr. Kethineni Vinod Kumar	100%	Nil	Syllabus completed; Linux shell scripting lab exercises verified.
2	Design and Analysis of Algorithms	II B.Tech II Sem	Er. M. Sateesh	95%	5%	NP-Completeness and approximation algorithms unit in progress; remedial classes scheduled.
3	Machine Learning & Deep Neural Networks	III B.Tech II Sem	Er. C. Harika	100%	Nil	Completed. Hands-on PyTorch model training and model question bank shared.
4	Compiler Design & Automata Theory	III B.Tech II Sem	Er. P. Rajesh	92%	8%	Code optimization and code generation modules concluding this week.

Dr. Kethineni Vinod Kumar

Head of Department

Department of Computer Science & Engineering

Dr. Sreenivas Prasad

IQAC Coordinator

Sanskriti School of Engineering

Principal

Institutional Endorsement

Sanskriti School of Engineering

